

VIETNAM NATIONAL UNIVERSITY-HO CHI MINH CITY

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

COMPUTER SCIENCE DEPARTMENT

Local Representation of Facial Features

Course: Pattern Recognition

Student Information:

23127004 - Le Nhat Khoi

23127165 - Nguyen Hai Dang

23127181 - Nguyen Tran Quoc Duy

Instructors:

Mr. Le Hoang Thai

Mr. Vo Thanh Lam

Mr. Nguyen Thanh Tinh

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CONTRIBUTION

Student ID	Full Name	Assigned Responsibilities	Rate
23127004	Le Nhat Khoi	- Core LBP Research: Read Chapter 1 & Section 2.1. Draft the explanations of the basic LBP algorithm, mathematical formulas, and the extraction process. - Input Analysis: Read the Augmented Features section of the paper. Draft Sections 5.1 & 5.2.1 explaining how the paper uses RGB, FFT Magnitude, and LBP as inputs. - The Link (Synthesis): Draft Section 6.1. Explain the theoretical derivation and methodological differences between the textbook (face recognition) and the paper (deepfake detection).	100%
23127165	Nguyen Hai Dang	- Extended Features: Read Sections 2.2 & 2.3. Draft the explanation of Gabor Features and the comparative analysis between spatial (LBP) and frequency (Gabor) domains. - Network Architecture: Read Chapters 3 & 4. Draft the sections on the Deep Learning transition, Binary Neural Network (BNN) theory, and its computational advantages. - Technical Evaluation: Draft Section 6.2. Evaluate the technical strengths (speed, accuracy) and limitations (video compression sensitivity, lack of temporal context) of the paper's method.	100%
23127181	Nguyen Tran Quoc Duy	- Practical Applications: Read Sections 2.1.2 & 2.1.3. Draft the explanation of facial histogram construction and traditional applications (Face Recognition/Verification). - Backbone & Output Analysis: Read Sections 5.2.2 to 5.2.4. Draft the explanations of the Adapter module, the BNext Backbone, and the Classification Head. - Extensions & Demo Plan: Draft Sections 6.3 & 6.4. Provide personal technical insights, suggest alternative applications, and design the step-by-step real-time demonstration plan.	100%

CHAPTER 0

OVERALL SUMMARY

This report examines the local representation of facial features, tracing the evolution from classical handcrafted descriptors to modern deep learning approaches, with a concrete case study in real-time deepfake detection.

Report Structure at a Glance

Chapter	Title	Key Content
1	Introduction	Introduce facial representation along with evolution of techniques and motivation for Deepfake detection
2	Local Representation Technique	LBP (basic, spatial, spatiotemporal, multi-resolution variants); Gabor filters (1-D, 2-D, filter bank); applications and comparative analysis
3	Deep Learning Era & Deepfake Detection	Limitation of handcrafted features and deep learning dominance; Deepfake detection problem formulation
4	Binary Neural Networks	Introduce a solution for lightweight deep model which the paper method relies on
5	Paper Methodology	Breakdown pipeline and modules along with explanation and analysis
6	Technique Analysis	Relationship to course theory; methodological differences; strengths & limitations; personal insights; extensibility; demonstration plan

Research Paper

- **Paper:** *“Faster Than Lies: Real-time Deepfake Detection using Binary Neural Networks”*
- **Authors:** Romeo Lanzino, Federico Fontana, Anxhelo Diko, Marco Raoul Marini, Luigi Cinque (Sapienza University of Rome)
- **Conference:** The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2024)
- **Rank:** A* (CORE ranking — top tier in Computer Vision)
- **Code:** https://github.com/fedelofer/binary_deepfake_detection

Chapter 1

Establishing that facial representation is the critical bottleneck in face analysis systems. A good descriptor must minimise intra-class variation, maximise inter-class discrimination, and be computationally efficient. Three historical eras are traced: Geometric (1970s), Holistic (1990s), and Local Descriptor (2000s). Real-time deepfake detection is introduced as the modern motivating application.

Chapter 2

Covering the two principal local descriptors. Local Binary Patterns (LBP) encode micro-texture by thresholding pixel neighbourhoods into binary codes. Variants include Circular and Uniform LBP for spatial flexibility, VLBP and LBP-TOP for spatiotemporal analysis, and Multi-scale / MB-LBP for multi-resolution coverage. Gabor filters extend the 1-D Gabor function to 2-D with orientation selectivity, forming a filter bank across multiple scales and orientations

Section 2.3

Comparing both descriptors: LBP is fast and illumination-robust but frequency-blind; Gabor captures rich spectral and directional structure at higher cost. Their complementarity — spatial texture (LBP) and frequency analysis (FFT/Gabor) — directly predicts the ablation finding in Chapter 5 that FFT + LBP together (91.60%) outperform either alone.

Chapter 3 & 4

Framing the transition to deep learning. As generative models produce increasingly convincing fakes, CNN-based detectors became dominant but most are too large for real-time use. Binary Neural Networks (BNNs) address this by quantising weights and activations to $\{-1, +1\}$, achieving a theoretical $58\times$ speedup and $32\times$ memory reduction via XNOR-POPCOUNT operations.

Chapter 5

Detailing the "Faster Than Lies" pipeline: RGB enriched with FFT magnitude (spectral fingerprint) and LBP (texture map) $\rightarrow$ Adapter compresses 5 channels to 3 $\rightarrow$ BNext binary backbone extracts features $\rightarrow$ full-precision classifier outputs real/fake. Ablation confirms FFT + LBP (91.60%) beats all other configurations and the baseline (90.35%).

Chapter 6

Bridging the paper to course theory. LBP's role as texture encoder maps directly to Chapter 2 theory; FFT parallels Gabor's frequency-domain insight. Key differences: LBP used as CNN channel instead of histogram, BNN backbone, Adapter layer, and deepfake detection instead of face recognition. Limitations (theoretical speedup, LBP parameters unspecified, no cross-dataset transfer) are identified, and extensions including Gabor channel augmentation and LBP-TOP for video deepfakes are proposed.

CHAPTER 1

INTRODUCTION

1.1 Facial Representation

Developing a robust face recognition system is based on two fundamental pillars: Facial Representation and Classifier Design.

While the classifier is responsible for making the final decision, the representation stage is arguably more critical. Its primary objective is to transform raw pixel data by computationally encoding into a meaningful set of features. The representation must minimize intra-class variations and while simultaneously maximizing extra-class variations.

1.2 Evolution of Techniques

However, finding a perfect feature set is challenging due to the immense variability in human facial appearance. Factors such as lighting conditions, camera pose, facial expressions, aging, and accessories (like glasses) create significant noise in the data.

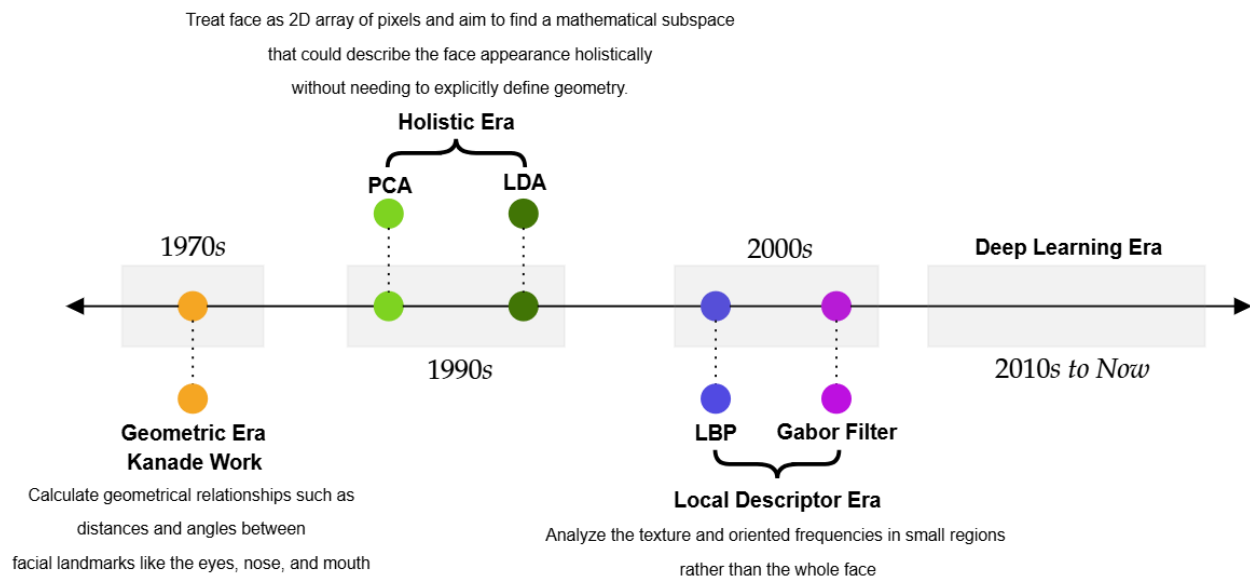


Figure 1: Technique development timeline

1.3 An Application to Deepfake Detection

The rapid progress of deep generative models now can produce highly realistic facial content known as DeepFake. However, they introduce serious concerns related to misinformation and privacy.

Traditional handcrafted descriptors such as Local Binary Patterns (LBP) and Gabor filters are effective at capturing local texture information; however, modern deepfakes often contain subtle artifacts that challenge shallow representations. Consequently, recent research has shifted toward deep learning-based solutions. In this context, the paper *“Faster Than Lies: Real-time Deepfake Detection using Binary Neural Networks”* proposes an efficient lightweight approach that enables accurate real-time deepfake detection on resource-constrained devices.

CHAPTER 2

LOCAL REPRESENTATION TECHNIQUE

2.1 Local Binary Patterns (LBP)

The Local Binary Pattern (LBP) is a highly popular local texture descriptor in computer vision. The core concept of LBP involves encoding the micro-structures of an image by comparing the intensity of a central pixel with its surrounding neighbors.

Due to its computational simplicity, high efficiency, and robustness to monotonic gray-scale transformations, LBP has been adopted in tasks such as face recognition, texture classification, and video analysis.

A typical LBP-based pipeline consists of four main stages: LBP Encoding, Histogram Construction, Feature Vector Formation and Similarity Measurement/Classification

2.1.1 Variation of LBP Operator

2.1.1.1 Basic LBP Operator

In the original version, the LBP is calculated within a 3×3 window. The LBP code at the center pixel is defined as:

$$LBP(x, y) = \sum_{p=0}^7 \text{sgn}(I_p - I_c) 2^p$$

where I_c be the gray value of the center pixel and I_p be the gray value of the p^{th} neighbour.

In fact, the final formula can be broken down into steps as the illustration:

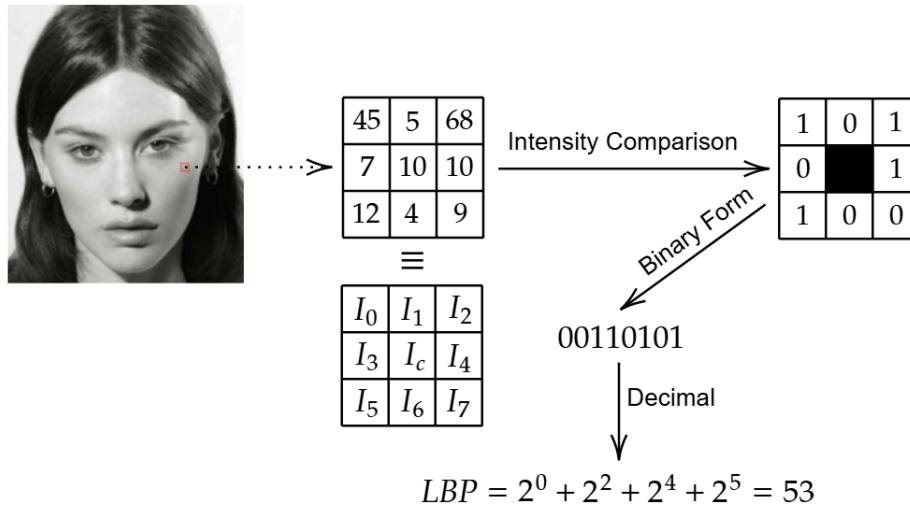


Figure 2: Step by step LBP illustration

After computing the LBP code for every pixel, we obtain an LBP label image. The histogram is defined as:

$$H_i = \sum_{x,y} \mathbf{1}\{LBP(x, y) = i\}, i = \overline{0, n-1}$$

where n is the number of different labels produced by the LBP operator

Finally, when the image patches whose histograms are to be compared have different sizes, the histograms must be normalized:

$$N_i = \frac{H_i}{\sum_{j=0}^{n-1} H_j}$$

2.1.1.2 Spatial LBP

The original LBP operator is limited in spatial support and produces high-dimensional histograms when generalized. Several spatial variants have been proposed to improve flexibility and robustness including:

- **Circular LBP:** Circular LBP is parameterized by:
 - P is the number of sampling points on the circle
 - R is the radius of the circle

As illustration, for a center pixel (x, y) , the neighbors are sampled on a circle of radius R :

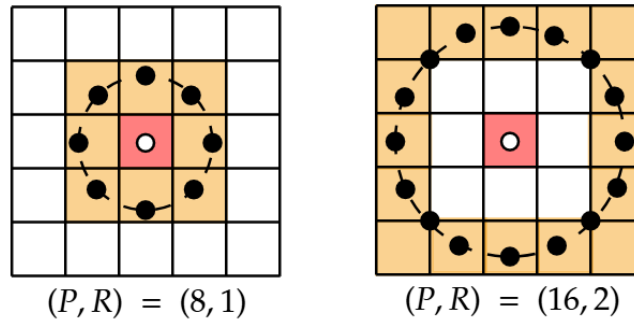


Figure 3: Circular LBP

In mathematical expression, the sampling points are located at:

$$(x_p, y_p) = (x + R \cos \frac{2\pi p}{P}, y - R \sin \frac{2\pi p}{P})$$

which requires Bilinear interpolation to estimate the gray value I_p

However, the number of possible labels grows exponentially 2^P which will lead to sparse histograms and unstable statistics.

- **Uniform LBP:** Uniform LBP is inspired by the fact that some binary patterns occur more frequently than others in texture images.

A binary pattern is called uniform if the circular bit string contains at most two transitions between 0 and 1. In mathematical expression, we can state:

$$\mathbf{b} \text{ is uniform} \iff U(\mathbf{b}) = \sum_{p=0}^{P-1} |\mathbf{b}_p - \mathbf{b}_{(p+1) \bmod P}| \leq 2$$

For example, you can calculate and identify whether the binary pattern is uniform or not:

$$\begin{array}{ll}
 \mathbf{b}^{(1)} = 01110000 \longrightarrow U(\mathbf{b}^{(1)}) = 2 & \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Uniform} \\
 \mathbf{b}^{(2)} = 11001111 \longrightarrow U(\mathbf{b}^{(2)}) = 2 & \\
 \mathbf{b}^{(3)} = 11001001 \longrightarrow U(\mathbf{b}^{(3)}) = 4 & \left. \begin{array}{l} \\ \\ \end{array} \right\} \text{Non-uniform} \\
 \mathbf{b}^{(4)} = 01010011 \longrightarrow U(\mathbf{b}^{(4)}) = 6 &
 \end{array}$$

Figure 4: Example illustration

In labeling strategy, each uniform pattern receives its own label and all non-uniform patterns are grouped into one bin. As a result, for $P = 8$, there are a total of 256 patterns of which 58 are uniform thus yielding to the total of 59 different labels.

In conclusion, the combination of these two extension, Uniform Circular LBP is commonly denoted $LBP_{P,R}^{u_2}$ where u_2 means uniform with at most 2 transitions.

2.1.1.3 Spatiotemporal LBP

In many real-world problems, especially facial expression analysis and dynamic texture recognition require modeling temporal variations. To address this limitation, LBP was extended into the spatiotemporal domain includes:

- **Volume LBP (VLBP):** This simply treats a video as a 3D volume in (X, Y, T) where:
 - X, Y are the spatial coordinates
 - T is the time coordinates (in other word, frame index)

Instead of analyzing a 2D neighborhood, VLBP analyzes a 3D neighborhood around each pixel in space-time. Here is the visualization:

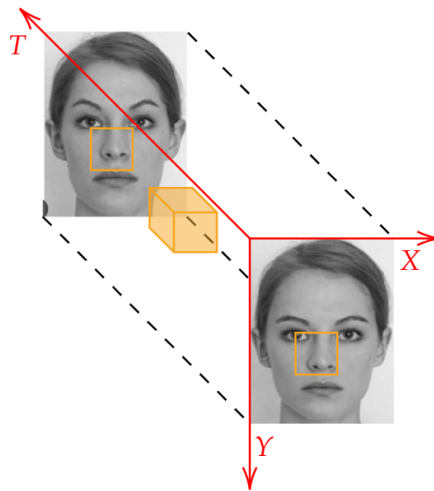


Figure 5: Volume LBP visualization

However, VLBP suffers from combinatorial explosion results in high-dimensional vectors and poor real-time suitability.

- **LBP-TOP (Three Orthogonal Planes):** To make VLBP more computationally efficient, LBP-TOP computes standard 2D LBP on three orthogonal planes through the video volume:

- **XY plane (Spatial Appearance):** $LBP_{P_{XY}, R_{XY}}^{u_2}(x, y, t) \rightarrow H_{XY} \rightarrow N_{XY}$
- **XT plane (Horizontal Motion):** $LBP_{P_{XT}, R_{XT}}^{u_2}(x, y, t) \rightarrow H_{XT} \rightarrow N_{XT}$
- **YT plane (Vertical Motion):** $LBP_{P_{YT}, R_{YT}}^{u_2}(x, y, t) \rightarrow H_{YT} \rightarrow N_{YT}$

where $H_{_}$ and $N_{_}$ are the histogram and normalized histogram of the corresponding plane, respectively. Then the final feature vector is generated by concatenating:

$$N = [N_{XY} \| N_{XT} \| N_{YT}]$$

2.1.1.4 Multi-resolution LBP

Single-scale LBP cannot simultaneously capture fine micro-textures or large micro-textures. Multi-resolution LBP has been proposed to fix this including:

- **Multi-scale LBP:** The approach is simple, Multiple LBP operators with different (P, R) will be applied. Each pixel is encoded at several scales:

$$LBP_{P_1, R_1}, LBP_{P_2, R_2}, \dots$$

Instead of building a joint histogram (which would be extremely sparse), separate histograms are computed for each scale and then concatenated.

- **Multi-scale Block LBP (MB-LBP):** MB-LBP extends LBP by comparing average intensities of blocks instead of individual pixels as illustration:

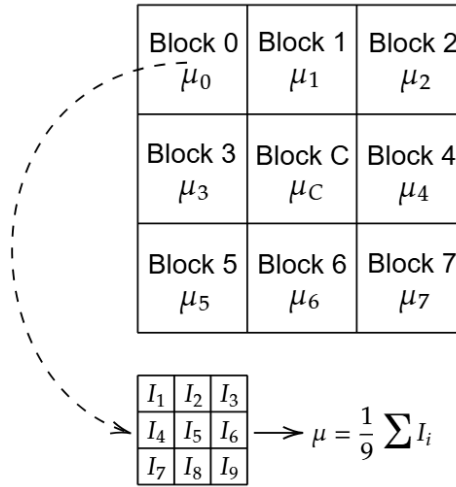


Figure 6: Multi-Scale Block LBP illustration

The procedure consists:

- Dividing neighborhood into sub-blocks.
- Computing mean intensity of each block.
- Applying the LBP comparison on block means.

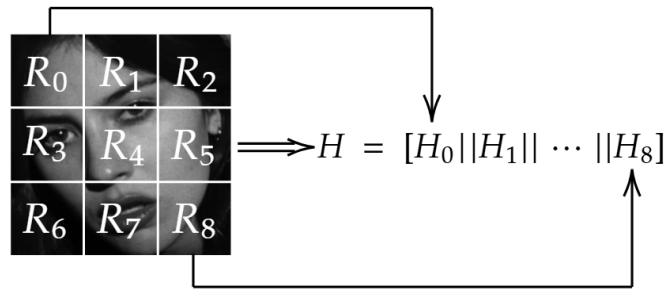
As a result, if we use 3×3 blocks, effective support may cover a 9×9 area.

2.1.2 Face Description with LBP

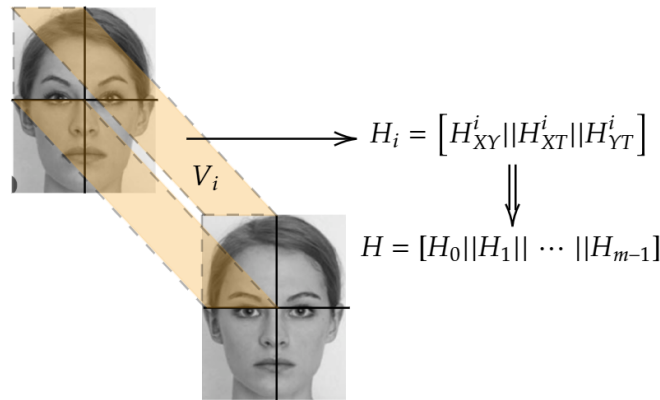
In classical texture classification, the occurrences of Local Binary Pattern (LBP) codes are typically accumulated into a single global histogram, and classification is performed by comparing histogram similarities.

However, applying this holistic strategy directly to facial images will make global histogram ignores spatial information.

Therefore, instead, the face image is divided into m local regions: $R_0, R_1, \dots, R_{m-1}$ which may do not have to be a rectangle and can be overlapped and varied in sizes. At each region, we then calculate LBP as well as their region histogram then simply concatenate them:



In the case of sequences of face, we perform the same method by divided into block volumes: $V_1, V_2, \dots, V_{m-1}$ with block histogram H_i make from plane histograms $H_i = [H_{XY}^i || H_{XT}^i || H_{YT}^i]$. Finally, we simply concatenate them as the way we do in static image:



2.1.3 Some Application with LBP Descriptor

Local Binary Pattern (LBP) descriptors have been successfully applied to a wide range of face analysis problems. This section will introduce some applications with LBP descriptor we present in the previous part.

2.1.3.1 Face Recognition

In face recognition, the goal is to identify the identity of a person from a facial image. After extracting the spatially enhanced LBP descriptor described in the previous section, recognition is typically performed using a distance-based classifier such as the nearest neighbor (NN) rule.

Let $\mathbf{x}$ and $\boldsymbol{\xi}$ denote two normalized enhanced histograms. The similarity between two faces is commonly measured using the Chi-square distance:

$$\chi^2(\mathbf{x}, \boldsymbol{\xi}) = \sum_{j=0}^{m-1} \sum_{i=0}^{n-1} \frac{(x_{i,j} - \xi_{i,j})^2}{x_{i,j} + \xi_{i,j}}$$

where i and j indexes the histogram bin and local facial region with m and n is the number of regions and LBP labels, respectively

Since different facial regions contribute unequally to recognition, a weighted version is used:

$$\chi_w^2(\mathbf{x}, \boldsymbol{\xi}) = \sum_{j=0}^{m-1} w_j \sum_{i=0}^{n-1} \frac{(x_{i,j} - \xi_{i,j})^2}{x_{i,j} + \xi_{i,j}}$$

where w_j is the importance weight of region j .

The identity of a probe face is then determined by the nearest neighbor rule:

$$\hat{y} = \arg \min_k \chi_w^2(\mathbf{x}, \mathbf{g}_k)$$

where $\mathbf{g}_k$ is the gallery descriptor of subject k .

There are also many other extensions and improvements including:

- **Tan–Triggs Preprocessing:** To further improve robustness under severe lighting, this was proposed that normalizes illumination before LBP extraction.
- **Multi-scale LBP with LDA:** Extracting multi-scale LBP histograms and projecting the concatenated feature vector into a discriminative subspace using LDA.
- **AdaBoost-based Region Selection:** AdaBoost is used to select the most informative facial regions and learn optimal region weights.

2.1.3.2 Face Verification

Face verification determines whether two facial images belong to the same person. Instead of nearest neighbor identification, verification is usually formulated as a binary hypothesis test.

In probabilistic LBP-based systems, local histograms are interpreted as empirical distributions. Given a client model $\mathcal{M}_c$ and a generic model $\mathcal{M}_g$, the decision is made using the log-likelihood ratio:

$$\Lambda(\mathbf{x}) = \log \frac{p(\mathbf{x} | \mathcal{M}_c)}{p(\mathbf{x} | \mathcal{M}_g)}, \text{ accept if } \Lambda(\mathbf{x}) > \tau$$

where τ is a predefined threshold.

2.1.3.3 Facial Expression Recognition

LBP and its spatiotemporal variants are widely used for facial expression analysis. Static LBP captures appearance information, while LBP-TOP encodes dynamic motion patterns from video sequences.

These descriptors are typically fed into classifiers such as:

- Support Vector Machine (SVM)
- k-Nearest Neighbor (kNN)
- Linear Discriminant Analysis (LDA)

2.1.3.4 Face Detection

LBP features are also effective for real-time face detection. In this task, compact LBP variants (e.g., $LBP_{4,1}$) and overlapping regions are often used to reduce feature dimensionality and improve statistical reliability on low-resolution images.

2.1.3.5 Other Face-Related Applications

Beyond the major tasks above, LBP-based descriptors have been successfully applied to:

- Gender recognition
- Age estimation
- Head pose estimation
- Iris recognition
- 3D face recognition

2.2 Gabor Feature

In visual analysis, we usually want to know two things at the same time: where a structure is located (spatial localization) and what frequency pattern it contains (frequency localization). However, there is a natural trade-off between these two goals:

- A pure sinusoid gives perfect frequency information, but it stretches infinitely in space, so it does not tell us where the pattern occurs.
- A Dirac delta impulse is perfectly localized in space, but it contains all frequencies equally, so it provides no frequency selectivity.

This limitation comes from the Fourier uncertainty principle. Dennis Gabor showed that a very good compromise can be achieved by multiplying a sinusoid with a Gaussian envelope.

2.2.1 1D Gabor Function

In one dimension, the complex Gabor function is defined as

$$\psi(t) = \exp\left(-\frac{(t - t_0)^2}{2\sigma^2}\right) \exp(j(2\pi f_0 t + \phi)).$$

where α is the bandwidth of Gaussian, t_0 is the time shift defining the time location of the Gaussian and f_0 is the frequency of the harmonic oscillations

2.2.2 From 1D to 2D

Images are two-dimensional signals, so the Gabor function must be extended to 2D. The additional parameter is orientation selectivity θ , since image structures such as edges have directions.

The 2D Gabor function becomes:

$$\psi(x, y) = \exp\left(-\left(\frac{f^2}{\gamma^2}x'^2 + \frac{f^2}{\eta^2}y'^2\right)\right) \exp(j2\pi f x'),$$

where the rotated coordinates are

$$x' = x \cos \theta + y \sin \theta, \quad y' = -x \sin \theta + y \cos \theta.$$

2.2.3 Gabor Feature Extraction

Given an input image $\xi(x, y)$ and a Gabor filter $\psi(x, y; f, \theta)$, the filter response is computed by convolution:

$$r_\xi(x, y; f, \theta) = \psi(x, y; f, \theta) * \xi(x, y).$$

At each spatial location, the complex response measures how strongly the local image pattern matches the filter's preferred frequency f and orientation θ .

However, a single Gabor filter captures only one frequency and one direction. To represent natural images effectively, a filter bank is constructed by sampling multiple frequencies and orientations.

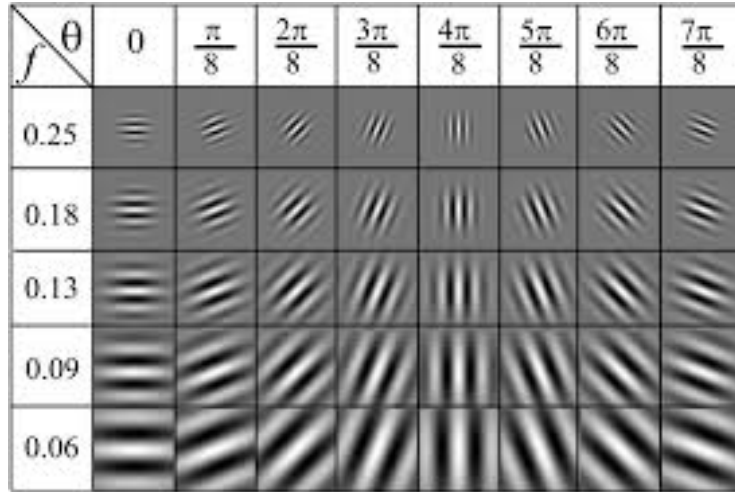


Figure 7: Gabor Bank visualization from [Research Gate](#)

- Frequencies are typically sampled on a logarithmic scale:

$$f_k = c^{-k} f_{\max}, \quad k = 0, \dots, m-1,$$

where $c > 1$ is the scaling factor.

- Orientations are uniformly spaced:

$$\theta_\ell = \frac{\ell\pi}{n}, \quad \ell = 0, \dots, n-1.$$

At a fixed spatial location (x_0, y_0) , the responses from all filters are collected to form a feature matrix:

$$G(x_0, y_0) = \begin{bmatrix} r(f_0, \theta_0) & \cdots & r(f_0, \theta_{n-1}) \\ \vdots & \ddots & \vdots \\ r(f_{m-1}, \theta_0) & \cdots & r(f_{m-1}, \theta_{n-1}) \end{bmatrix}.$$

Rows correspond to different scales (frequencies) and columns correspond to different orientations. This local descriptor is known as a *Gabor jet*.

For machine learning applications, the matrix is typically flattened into a single feature vector

$$g(x_0, y_0) = [r(f_0, \theta_0), \cdots, r(f_{m-1}, \theta_{n-1})]$$

2.2.4 Application

Once Gabor features have been extracted and statistically modeled, they can be used in several face analysis tasks. The common theme is to treat the Gabor jet at each location as a discriminative local descriptor and then apply either probabilistic models or distance-based matching.

2.2.4.1 Facial Landmark Detection

Let each facial landmark type (left eye, right eye, nose tip, etc.) be treated as a separate class. Given an image location (x, y) with Gabor feature vector $\mathbf{g}(x, y)$, we evaluate how likely this location belongs to each landmark class using

$$p(\mathbf{g}(x, y) \mid \text{class}).$$

The detected landmark position is the location with the highest likelihood.

The full pipeline is illustrated in the following image:

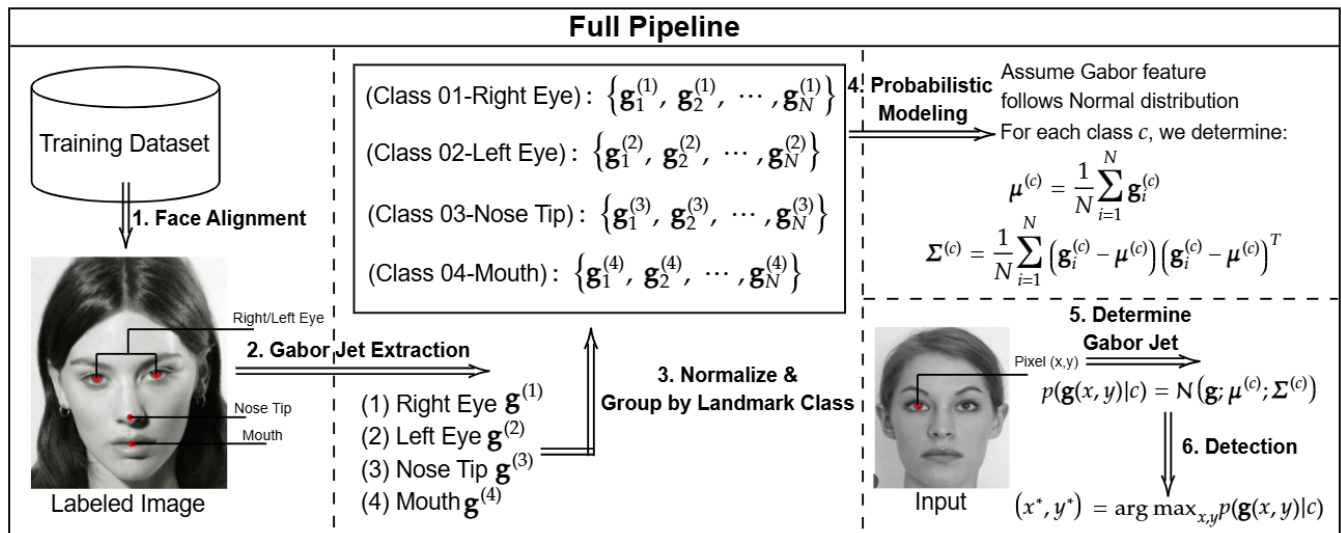


Figure 8: Full Pipeline of Facial Landmark Detection

2.2.4.2 Face Recognition (Identification)

After detecting and aligning the face, Gabor jets are extracted at a set of stable facial landmarks. These local descriptors are concatenated into a global face feature vector.

Recognition is then performed by comparing the probe face against a gallery of known identities using Distance Measures or Likelihood under class models.

2.3 Final Conclusion

Local feature descriptors are fundamental to robust face representation under variations in illumination, pose, and imaging conditions. Among them, Local Binary Patterns (LBP) and Gabor-based features remain two of the most influential approaches.

Property	LBP	Gabor Filter
Primary information	Local texture / intensity ordering	Frequency & orientation content
Computational cost	Very low (integer ops)	High (complex convolutions)
Illumination robustness	High (rank-order based)	Moderate (normalisation-dependent)
Frequency selectivity	None	Strong (multi-scale bank)
Spatial locality	Strong (3×3 to custom P, R)	Strong (Gaussian envelope)
Typical use	Real-time texture classification	Precision face analysis

LBP offers fast and illumination-robust texture encoding by quantizing local intensity ordering. Its main limitation is sensitivity to noise and flat regions, which can lead to unstable codes. Numerous variants—such as LTP, Dominant LBP, and LDP—have been proposed to improve robustness. LBP has also inspired related descriptors like WLD and LPQ, the latter showing strong blur robustness through local phase quantization.

In contrast, **Gabor filters** provide rich frequency and orientation selectivity, making them highly effective for detailed facial analysis. Many studies demonstrate that combining LBP and Gabor features yields superior recognition performance, although at the cost of very high feature dimensionality that often requires techniques such as Fisher Discriminant Analysis.

Overall, LBP emphasizes computational efficiency and illumination invariance, whereas Gabor features capture discriminative spectral structure. Their complementary nature has driven the development of hybrid representations that bridge classical handcrafted descriptors and modern efficient learning-based models.

This complementarity is not merely theoretical: the research paper studied in Chapters 5–6 directly exploits it by combining an **LBP channel** (spatial texture) and a **FFT magnitude channel** (global spectral fingerprint—a computationally efficient proxy for the Gabor bank) as input to a Binary Neural Network. Their ablation study confirms that **FFT + LBP together (91.60%) outperform either alone** and the RGB baseline (90.35%), validating the complementarity prediction from this chapter.

There is also a deeper conceptual resonance worth noting: LBP binarises pixel-neighbourhood relationships using the sign function; Binary Neural Networks binarise the *entire network's* weights and activations using the same sign function. This echo - from the classical binary texture descriptor of the early 2000s to modern binary deep networks - forms the unifying thread of this report.

CHAPTER 3

DEEP LEARNING ERA & DEEPPAKE DETECTION

3.1 Deep Learning Transition

Motivation: The limitations of handcrafted facial descriptors have motivated the research community to shift toward deep learning-based solutions. Traditional features such as LBP or Gabor are efficient and interpretable, but they often struggle to capture highly complex and nonlinear visual patterns present in modern face analysis tasks.

Dominance of Deep Learning: With the rapid growth of computational power and large-scale datasets, Convolutional Neural Networks (CNNs) have become the dominant paradigm for learning robust facial representations directly from data.

An Effort of Building Lightweight Deep Model: However, many state-of-the-art approaches rely on large and computationally expensive architectures, which limits their deployment in real-time or resource-constrained environments. This challenge motivates recent research toward lightweight yet accurate deep learning models.

3.2 Deepfake Detection Problem & Previous Approaches

Formally, given an input image $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$, the goal of deepfake detection is to learn a classifier

$$f_{\theta}(\mathbf{x}) \rightarrow \{0, 1\}$$

where 0 and 1 denote real and fake images, respectively.

The classifier f_{θ} is typically implemented using deep neural networks parameterized by θ , and trained on a labeled dataset

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N,$$

where $y_i \in \{0, 1\}$ indicates whether the image is authentic or manipulated. The learning objective is commonly formulated as a binary classification problem using the cross-entropy loss. There are several previous approaches to this problem includes:

- **Artifact-based Methods:** Early detectors focused on hand-crafted visual artifacts such as frequency irregularities, color inconsistencies, or physiological cues. While computationally efficient, these methods often overfit to specific manipulation pipelines and generalize poorly to unseen deepfakes.
- **CNN-based Approaches:** To enhance robustness, later works adopted convolutional neural networks that automatically learn discriminative features from raw images. Their ability to capture complex spatial patterns has made CNN-based detectors the dominant approach in modern deepfake detection.

Limitation Despite strong performance, most high-accuracy detectors rely on large backbone networks with millions of parameters and high floating-point computational cost (FLOPs) making it difficult deploying real-time system

→ We need to find a more efficient and real-time approach

CHAPTER 4

BINARY NEUTRAL NETWORK

These challenges motivate the exploration of lightweight deep learning models that preserve discriminative power while reducing computational overhead. Among various model compression strategies, Binary Neural Networks (BNNs) have emerged as an effective approach for building computationally efficient deep models. The key idea is to quantize both network weights and activations to binary values, typically in $\{-1, +1\}$, thereby replacing most floating-point multiplications with inexpensive bit-wise operations.

4.1 From Full-Precision CNNs to BNNs

In a standard convolutional neural network, the output feature map of a convolution layer is computed as

$$\mathbf{Y} = \mathbf{A}_r \otimes \mathbf{W}_r,$$

where $\mathbf{A}_r$ and $\mathbf{W}_r$ denote real-valued activations and weights.

Binary Neural Networks (BNNs) approximate this computation by binarizing both operands using the sign function:

$$\mathbf{A}_b = \text{sgn}(\mathbf{A}_r), \quad \mathbf{W}_b = \text{sgn}(\mathbf{W}_r).$$

However, direct binarization causes significant information loss. The real-valued tensors are approximated with scaling factors:

$$\mathbf{A}_r \approx \alpha \mathbf{A}_b, \quad \mathbf{W}_r \approx \beta \mathbf{W}_b,$$

where

$$\alpha = \frac{1}{n} \|\mathbf{A}_r\|_1, \quad \beta = \frac{1}{n} \|\mathbf{W}_r\|_1.$$

Therefore, the convolution can be approximated as

$$\mathbf{Y} = \mathbf{A}_r \otimes \mathbf{W}_r \approx (\alpha \mathbf{A}_b) \otimes (\beta \mathbf{W}_b) = (\alpha\beta) (\mathbf{A}_b \circledast \mathbf{W}_b),$$

where $\circledast$ denotes bit-wise XNOR followed by population count (POPCOUNT).

4.2 Computational Advantages

The binarization process yields significant efficiency gains:

- multiplications $\rightarrow$ bit-wise XNOR operations,
- floating-point accumulations $\rightarrow$ integer POPCOUNT,
- memory usage reduced by up to 32 times.

As a result, the theoretical computational cost of binary convolution is dramatically lower than that of full-precision CNNs, making BNNs particularly attractive for real-time applications.

CHAPTER 5

PAPER METHODOLOGY

5.1 Paper's Position

Building upon these developments, the paper *“Faster Than Lies: Real-time Deepfake Detection using Binary Neural Networks”* explores the use of a modern binary backbone (BNext) for efficient deepfake detection. The core idea is to combine:

- The efficiency of Binary Neural Networks,
- The representation power of deep CNN backbones,
- The complementary handcrafted cues (FFT and LBP).

By doing so, the authors aim to bridge the gap between **high detection accuracy** and **real-time deployability**. The method achieves state-of-the-art accuracy on three benchmark datasets (COCOFake, DFFD, CIFAKE) with **up to 20× reduction in FLOPs** during inference compared to existing approaches. Figure 1 of the paper shows BNext-T (0.89 GFLOPs) achieving 99.25% accuracy on COCOFake — matching or beating models requiring up to 8.56 GFLOPs.

5.2 Methodology & Analysis

The proposed framework aims to classify an input RGB image as either *real* or *fake*. As illustration, pipeline consists of **three** main stages:

- (i) Augmented feature construction
- (ii) Channel adaptation via an Adapter module
- (iii) Feature extraction using a binary backbone followed by Classifier

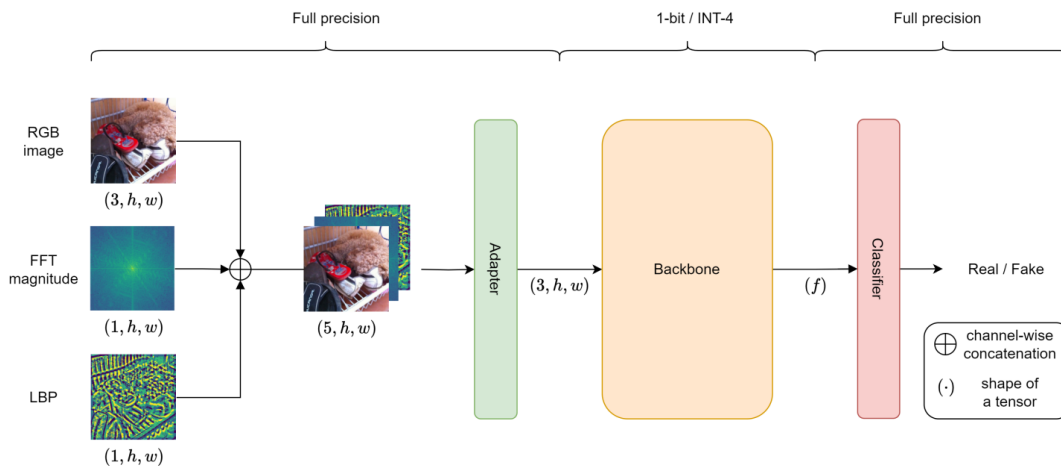


Figure 9: Overall Pipeline

5.2.1 Augmented Features

Given an RGB image $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$ is added with two channels with FFT magnitude and LBP.

- **FFT magnitude channel:** The RGB image is first converted to grayscale and transformed into the frequency domain using the 2D Fast Fourier Transform (FFT):

$$\mathbf{F} = |\mathcal{F}(\mathbf{x}_{gray})|.$$

To enhance the visibility of high-frequency artifacts, a logarithmic scaling is applied:

$$\mathbf{F} := \log(1 + \mathbf{F}).$$

This channel captures spectral anomalies—such as periodic patterns and frequency irregularities—often introduced by generative models. GANs and diffusion models leave characteristic frequency-domain artefacts (e.g., upsampling and smoothing effects) that are difficult to observe spatially. Hence, the FFT magnitude serves as a global spectral fingerprint of potential manipulation.

- **LBP channel:** Local Binary Pattern (LBP) is then computed in a circular neighborhood with $P = 8$ sampling points and radius $R = 1$: $LBP_{8,1}$ which is just the simple LBP operation

The final augmented input concatenates all channels:

$$\mathbf{x}_{\text{aug}} = [\mathbf{x} \parallel F \parallel \text{LBP}] \in \mathbb{R}^{5 \times H \times W}$$

The ablation study from the original paper confirms the complementary nature of these features. Using either FFT or LBP alone improves performance over the RGB baseline, whereas their combination yields the best results.

Table 1: Ablation Result in COCOFake Dataset

Configuration	Accuracy (%)
Baseline (RGB only)	90.35
Magnitude	82.36
FFT	88.18
LBP	88.42
Magnitude + FFT	81.20
Magnitude + LBP	81.67
FFT + LBP (chosen)	91.60
Magnitude + FFT + LBP	81.56

This suggests that frequency-domain and local texture cues capture different aspects of forgery artifacts and jointly provide a more discriminative representation.

5.2.2 Adapter Module

Since the backbone expects 3-channel input, an Adapter layer is introduced to fuse the augmented features. The Adapter is implemented as a learnable convolution:

$$\mathbf{x}_{adapt} = \phi(\mathbf{W}_a \otimes \mathbf{x}_{aug}),$$

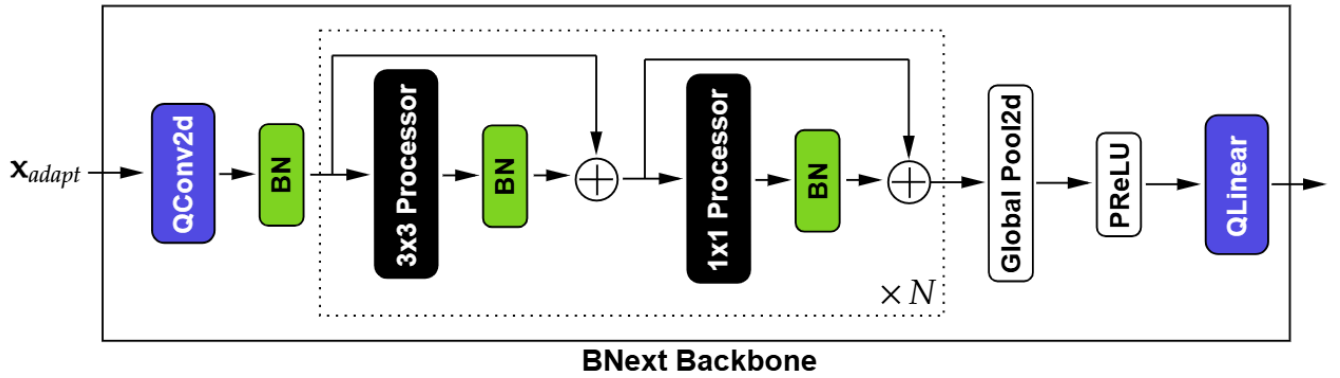
where $\mathbf{W}_a$ denotes the Adapter weights and $\phi(\cdot)$ is a nonlinear activation. This operation projects the 5-channel tensor back to

$$\mathbf{x}_{adapt} \in \mathbb{R}^{3 \times H \times W},$$

while preserving the most informative cues from the augmented representation.

5.2.3 Binary Backbone

To enable real-time inference, the proposed framework adopts a **Binary Neural Network (BNN) backbone**. Specifically, the paper employs the BNext architecture which significantly reduces memory footprint and computational cost while preserving discriminative capability for deepfake detection:



In BNext, each convolution operation is **fully binary**, and other operations are **quantised to INT-4**, resulting in much more efficient networks in terms of operations. Formally, the binary convolution approximates full-precision computation as follows. Starting from the standard convolution $Y = A_r \otimes W_r$, BNext binarises both weights and activations using the sign function:

$$A_b = \text{sign}(A_r), \quad W_b = \text{sign}(W_r)$$

where $\text{sign}(x_r) = +1$ if $x_r \geq 0$, else -1 . Following XNOR-Net, a channel-wise scaling factor α is introduced to reduce quantisation error. The binary convolution output becomes:

$$Y \approx (A_b \circledast W_b) \odot \alpha$$

where $\circledast$ denotes **XNOR-POPCOUNT** bit-wise operations and $\odot$ is element-wise multiplication. This yields a theoretical **58× inference speedup and 32× memory reduction** compared to full-precision CNNs.

The backbone uses a **binary convolution module with full-precision skip-connections** and a branch with precision INT-4 to facilitate information propagation and alleviate possible bottlenecks. Three model sizes are evaluated:

- **BNext-T** (Tiny): 0.89 GFLOPs, 29.8M parameters
- **BNext-S** (Small): 1.91 GFLOPs, 67.1M parameters
- **BNext-M** (Medium): 3.39 GFLOPs, 133M parameters

For reference, ResNet-50 requires 4.8 GFLOPs and ViT-B/32 requires 8.56 GFLOPs. Even BNext-T achieves **99.25% accuracy on COCOFake at just 0.89 GFLOPs** with a trainable backbone - demonstrating that the BNN efficiency gains do not come at the cost of detection quality. All backbones are pre-trained on ImageNet (1.2M images).

Two evaluation configurations are tested: a **frozen backbone** (conserves training resources, assesses zero-shot transfer) and a **trainable backbone** (maximises accuracy). The results across three datasets confirm that the trainable BNext consistently matches or outperforms larger full-precision models at a fraction of the computational cost.

5.2.4 Classification Head

Let $\mathbf{f} \in \{-1, +1\}^d$ denote the features extracted by the binary backbone. A full-precision linear classifier produces the logits:

$$\mathbf{z} = \mathbf{W}_c \mathbf{f} + \mathbf{b}_c,$$

followed by a sigmoid function to obtain the final prediction

$$\hat{y} = \sigma(\mathbf{z}) \in [0, 1],$$

where $\hat{y}$ indicates the probability that the input image is fake. The model is trained with binary cross-entropy loss using the AdamW optimiser ($\text{lr}=10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\text{weight_decay}=10^{-2}$) with a learning rate scheduler that reduces to 10^{-5} by the end of epoch 5.

The full-precision classification head is kept in floating point to ensure precise decision boundaries, since binarising the final logit layer would degrade accuracy without meaningful efficiency gains (it represents a negligible fraction of total operations).

CHAPTER 6

TECHNIQUE ANALYSIS & DIFFERENCE

6.1 Technical Relationship & Methodological Differences

6.1.1 Derivation from the Book Chapter

The paper builds directly upon the LBP theory covered in Chapter 2. Specifically, the authors employ LBP as a texture feature channel to augment the RGB input of a convolutional neural network. The motivation for this choice is rooted in the same fundamental insight presented in Section 2.1: LBP encodes local micro-structural patterns by comparing each pixel with its neighbours and is robust to monotonic illumination changes — properties that are equally valuable when the goal is detecting subtle texture anomalies introduced by generative models rather than identifying individuals.

The connection is conceptually direct. In the classical LBP face recognition pipeline (Section 2.1.3.1), LBP histograms serve as the primary feature for a nearest-neighbour classifier. In the paper, LBP serves as an auxiliary texture channel feeding into a deep neural network classifier. The descriptor remains identical in formulation: pixel-wise neighbourhood comparison producing binary codes that are assembled into a feature map (here used as an additional image channel rather than a histogram). The paper’s rationale for choosing LBP — that deepfakes typically struggle to maintain accuracy in facial texture regions — mirrors exactly the discriminative power that makes LBP useful for face recognition.

Beyond LBP, the paper supplements texture information with FFT magnitude, which captures frequency-domain anomalies introduced by generative processes. This resonates with the Gabor filter framework in Section 2.2: both FFT and Gabor filters operate in the frequency domain to capture structured patterns not visible in the spatial domain alone. Gabor filters provide localised frequency analysis at multiple scales and orientations, while the FFT magnitude provides a global frequency profile of the entire image. The authors thus implicitly leverage the same theoretical insight — that frequency-domain features reveal manipulation traces — but implement it via a computationally simpler global FFT rather than a spatial filter bank.

6.1.2 Methodological Differences

While the conceptual foundations are shared, the methodological approach differs fundamentally from classical local descriptor pipelines in several important ways:

- **Integration paradigm:** classical LBP-based systems use hand-crafted histogram features as the sole input to a shallow classifier (e.g., nearest-neighbour or SVM). The paper uses LBP as a pre-computed feature channel fed into a deep neural network, allowing the network to learn how to combine texture, frequency, and colour information in a data-driven manner.
- **Binary Neural Network backbone:** the most novel methodological contribution is the use of BNNs (specifically BNext), which quantise weights and activations to 1-bit values. This replaces expensive floating-point multiply-accumulate operations with XNOR and POP-COUNT bit-wise operations, yielding a theoretical $58\times$ speedup and $32\times$ memory reduction.

No classical LBP system employs network binarisation; the concept is entirely specific to deep learning.

- **Adapter layer:** an additional convolutional Adapter layer is introduced to compress the five-channel augmented input (RGB + FFT + LBP) back to three channels compatible with the pre-trained BNext backbone. This design pattern has no analogue in classical descriptor pipelines.
- **Task domain:** classical LBP descriptors were designed for face recognition and expression analysis using genuine images. The paper applies LBP to deepfake detection — a fundamentally different task where the goal is binary classification of real vs. generated images rather than identity or expression matching.

6.2 Technical Evaluation: Strengths, Limitations, and Applicability

6.2.1 Technical Strengths

The paper demonstrates several significant technical strengths:

- **Efficiency without accuracy sacrifice:** the BNN backbone requires only 0.89 GFLOPs for BNext-T compared to 4.8 GFLOPs for ResNet-50, while achieving substantially higher accuracy on COCOFake (99.25% vs. 90.31%). This is a remarkable result demonstrating that reduced precision does not necessarily degrade detection quality.
- **Principled feature augmentation:** the choice of LBP and FFT magnitude as auxiliary channels is well-motivated. Both address the shortcomings of RGB features alone: LBP captures local texture anomalies (spatial domain) while FFT magnitude exposes spectral artefacts (frequency domain). The ablation study confirms their complementarity — using FFT and LBP together achieves 91.60% accuracy, outperforming the baseline (90.35%) and all single-channel configurations.
- **Comprehensive evaluation:** the method is validated on three datasets spanning different generation methods (diffusion models in COCOFake and CIFAKE; GANs in DFFD) and image types (natural scenes and faces). Competitive or superior results across all three demonstrate reasonable generalisability.
- **Open-source reproducibility:** the code is publicly available, enabling direct replication and extension of the results.

6.2.2 Limitations and Assumptions

Several important limitations should be acknowledged:

- **Theoretical vs. practical speedup:** the paper acknowledges that BNN speedups are theoretical. Realising the $58\times$ inference speedup requires specialised hardware or frameworks that support XNOR/POPCOUNT operations. On standard GPU or CPU platforms without BNN-aware acceleration, the actual speedup may be considerably smaller.

- **Pre-training dataset constraint:** the backbone is pre-trained on ImageNet (1.2M images), while competing models pre-trained on much larger datasets (e.g., LAION-2B with 2 billion images) achieve higher accuracy on COCOFake. The gap narrows significantly with a trainable backbone, but the dependence on pre-training quality remains a constraint.
- **Limited cross-dataset generalisation analysis:** while the paper tests on three datasets, it does not explicitly evaluate cross-dataset transfer (train on one, test on another). This is a known weakness in deepfake detection research where models often fail to generalise to unseen generation techniques.
- **Sobel filter finding:** the ablation study shows that the Sobel edge filter does not improve performance, suggesting that the model’s native feature learning already captures edge information adequately. This also implies that not all intuitive feature augmentations are beneficial — careful ablation is essential.
- **LBP configuration:** the paper does not specify which LBP variant (basic, uniform, circular) or parameters (P, R) were used, limiting reproducibility of the exact LBP channel and making it difficult to understand whether the choice was optimised or arbitrary.

6.3 Personal technical insights and extensibility

6.3.1 Personal Insights

From a pattern recognition perspective, the paper is notable for demonstrating that classical, hand-crafted local descriptors — specifically LBP, which dates to the early 2000s — retain practical value even in modern deep learning pipelines. Rather than discarding pre-deep-learning feature engineering, the authors repurpose LBP as a domain-specific inductive bias: instead of asking the network to learn texture sensitivity from scratch, they directly encode it via LBP and provide it as a structured input channel.

The ablation result is particularly instructive: LBP alone (88.42%) is slightly more useful than FFT alone (88.18%), and together they synergise to 91.60% — exceeding the baseline by over one point. This confirms the theoretical prediction from Chapter 2 that spatial texture features (LBP) and frequency features (Gabor/FFT) are complementary rather than redundant. The Sobel filter’s failure to contribute further supports the finding that edge information is a subset of what LBP already captures, since LBP inherently encodes edge-like micro-patterns through its neighbourhood comparison.

The choice of BNNs is also conceptually elegant: just as LBP binarises the relationship between a central pixel and its neighbours (a core concept from Section 2.1), BNNs binarise the weights and activations of the entire network. There is thus a pleasing theoretical resonance between the classical binary descriptor and the binary network architecture.

6.3.2 Extensibility and Alternative Applications

Several promising research directions extend naturally from this work:

- **Gabor channel augmentation:** replacing or supplementing the FFT magnitude channel with a Gabor filter bank channel (as studied in Section 2.2) could provide richer, multi-scale,

multi-orientation frequency information while remaining computationally feasible. A filter bank with, say, 4 frequencies $\times$ 4 orientations could expose directional artefacts introduced by generative models more precisely than a global FFT.

- **Spatiotemporal extension for video deepfakes:** applying LBP-TOP (Section 2.1.1.3) as a channel augmentation for video-based deepfake detection would directly extend this framework to the temporal domain. LBP-TOP encodes motion patterns that deepfakes may fail to reproduce accurately, particularly around facial dynamics such as blinking and speaking.
- **Multi-scale LBP channels:** using multiple LBP channels at different (P, R) scales as input augmentation could capture micro-textures at varying granularities, potentially improving detection of both coarse GAN artefacts and fine diffusion model inconsistencies.
- **Real-time mobile deployment:** the efficiency advantage of BNNs makes this architecture a strong candidate for on-device deepfake screening on mobile phones — a directly relevant application given that deepfakes primarily spread through social media platforms accessed on mobile devices.

6.4 Demonstration Plan

To provide a convincing and reproducible demonstration of the core concepts at the final presentation, the following implementation steps are planned:

1. **Dataset Preparation:** sample real and manipulated face images from publicly available datasets such as FaceForensics++ or Celeb-DF. These will provide visually comparable real/fake pairs that clearly illustrate the detection challenge.
2. **LBP Feature Visualisation:** implement the LBP operator (e.g., using *skimage.feature.local_binary_pattern*) on both real and fake face images. Side-by-side visualisation of the RGB frame alongside its LBP feature map will make the texture encoding tangible and show qualitative differences between genuine and manipulated regions.
3. **FFT Magnitude Comparison:** compute and display the FFT magnitude spectra of real vs. fake images to demonstrate the frequency-domain anomalies that the paper's FFT channel is designed to capture.
4. **Real-time Pipeline Demo:** integrate face bounding-box detection (OpenCV) with LBP feature extraction in a live webcam or pre-recorded video stream. The output frame will display the LBP feature map overlay alongside an FPS counter, demonstrating the computational efficiency that motivates the paper's efficiency-first design philosophy.
5. **Ablation Visualisation:** reproduce the key finding of Table 4 (from the paper) in a bar chart showing accuracy for each feature combination (Baseline, LBP only, FFT only, FFT + LBP), illustrating the synergy between the two channels.

References

- [1] Lanzino, R., Fontana, F., Diko, A., Marini, M. R., & Cinque, L. (2024). *Faster Than Lies: Real-time Deepfake Detection using Binary Neural Networks*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2024)* (pp. 3771–3780).
- [2] Ojala, T., Pietikäinen, M., & Mäenpää, T. (2002). Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(7), 971–987.
- [3] Ahonen, T., Hadid, A., & Pietikäinen, M. (2006). Face description with local binary patterns: Application to face recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(12), 2037–2041.
- [4] Zhao, G., & Pietikäinen, M. (2007). Dynamic texture recognition using local binary patterns with an application to facial expressions. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(6), 915–928.
- [5] Daugman, J. G. (1985). Uncertainty relation for resolution in space, spatial frequency, and orientation optimized by two-dimensional visual cortical filters. *Journal of the Optical Society of America A*, 2(7), 1160–1169.
- [6] Wiskott, L., Fellous, J. M., Krüger, N., & von der Malsburg, C. (1997). Face recognition by elastic bunch graph matching. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 19(7), 775–779.
- [7] Courbariaux, M., Hubara, I., Soudry, D., El-Yaniv, R., & Bengio, Y. (2016). Binarized neural networks: Training deep neural networks with weights and activations constrained to +1 or -1. *arXiv preprint arXiv:1602.02830*.
- [8] Guo, N., Bethge, J., Meinel, C., & Yang, H. (2022). Join the high accuracy club on ImageNet with a binary neural network ticket. *arXiv preprint arXiv:2211.12933*.